

DEENA VARGHESE

AI Engineer

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PROFILE SUMMARY

AI Engineer with over 4 years of hands-on experience in building machine learning, deep learning, and generative AI solutions. Skilled in developing LLM-based chatbots, computer vision pipelines, and end-to-end AI products from data ingestion to deployment. Currently working at Ospyn Technologies, contributing to enterprise-grade AI products using LangChain, Ollama, and RAG architectures.

CORE SKILLS

Programming & ML: Python, TensorFlow, PyTorch, Keras, scikit-learn, NumPy, Pandas

LLMs & GenAI: LangChain, Ollama, Hugging Face Transformers, Prompt Engineering, RAG

Computer Vision & OCR: OpenCV, EasyOCR, Tesseract, YOLO

Web & API Development: Flask, FastAPI, REST APIs, Unicorn

MLOps & Deployment: Docker, CI/CD Pipelines, Model Serving, GitLab

Data Tools: Power BI, SQL, JSON, Qdrant, FAISS

Soft Skills: Analytical Thinking, Team Collaboration, Problem Solving, Project Coordination

EXPERIENCE

Ospyn Technologies Limited | AI Engineer | June 2024 – Present | Kochi, Kerala

- Designed and deployed RAG-based conversational chatbots integrated with LLMs for intelligent document interaction.
- Built document forgery detection systems using OpenCV, including orientation correction and feature matching.
- Trained and fine-tuned YOLO models for detecting and labeling document regions of interest.
- Developed OCR data extraction pipelines for structured and semi-structured scanned documents.
- Implemented semantic similarity and keyword-based research paper analysis using Ollama and embedding models.
- Worked with cross-functional teams to integrate AI modules ensuring performance, scalability, and reliability.

ThoughtRipples Technologies | Junior Data Scientist | Jan 2022 – June 2024 | Kochi, Kerala

- Built and deployed ML models (classification, regression, clustering) for analytics and forecasting.
- Conducted EDA and data preprocessing for large datasets using Pandas and scikit-learn.
- Created data visualizations and dashboards for actionable business insights using Power BI and Matplotlib.
- Assisted in model deployment and monitoring, improving model performance in production.
- Collaborated across teams to convert business requirements into data-driven solutions.

ThoughtRipples Technologies | Project Coordinator | Jan 2017 – Dec 2021 | Kochi, Kerala

- Managed project planning, scheduling, and cross-team communication.
- Monitored deliverables, maintained documentation, and supported risk management.

EDUCATION

Bachelor of Engineering (B.E.) – Electrical and Electronics Engineering

Anna University, CSI College of Engineering, Ooty

CERTIFICATION

Advanced Certification in Data Science and Artificial Intelligence – IIT Madras & Intellipaath